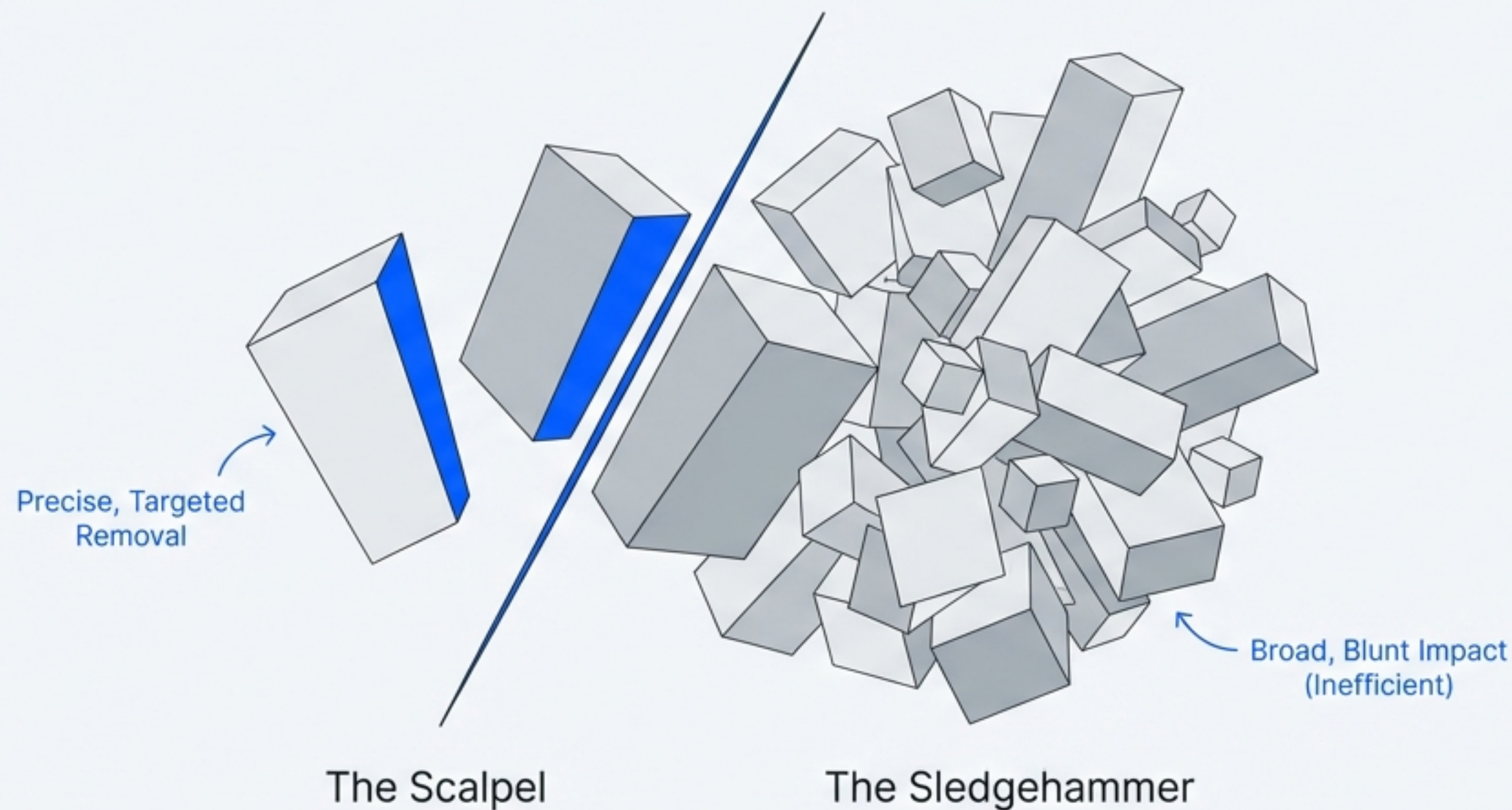


The Scalpel and the Sledgehammer: A New Era of Machine Unlearning

Introducing a precise, efficient, and certified framework for removing features and labels from trained models



Based on the work "Machine Unlearning of Features and Labels" by Warnecke, Pirch, Wressnegger, and Rieck (NDSS Symposium 2023).

What a model learns, it can leak.

Large-scale models, particularly in NLP, can inadvertently memorize and expose sensitive information from their training data. This creates significant privacy and security risks.

The paper cites a real-world example where a text completion system was found to contain users' credit card numbers.



My credit card number is... |

4111 1111 2222 3333

The need to forget is a fundamental challenge for modern AI.

Beyond isolated data points, we need mechanisms to correct models for various reasons. Retraining from scratch is often not a viable solution.



Regulatory Compliance

The “Right to be Forgotten” under regulations like GDPR requires the removal of personal data from all systems, including trained models.



Ethical Correction

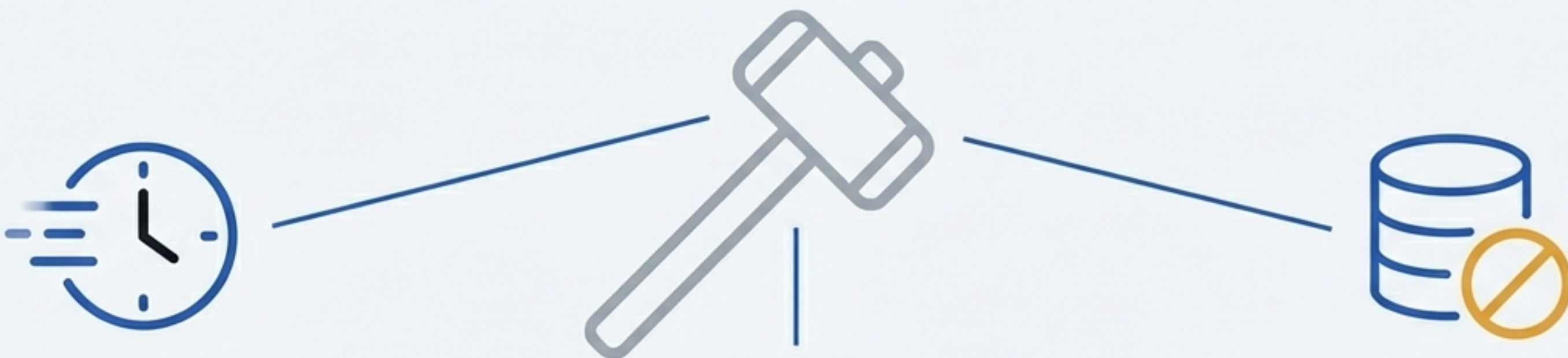
Models can memorize and replicate toxic or biased content, requiring immediate removal to prevent harm. The paper cites Microsoft’s “Tay” chatbot as an example.



Security & Robustness

Malicious actors can intentionally poison training data. Repairing this damage is critical for model integrity.

The Sledgehammer: Retraining is thorough but prohibitively expensive.



Massive Compute Cost:

Reproducing the entire training process consumes significant time and resources, especially for large models.

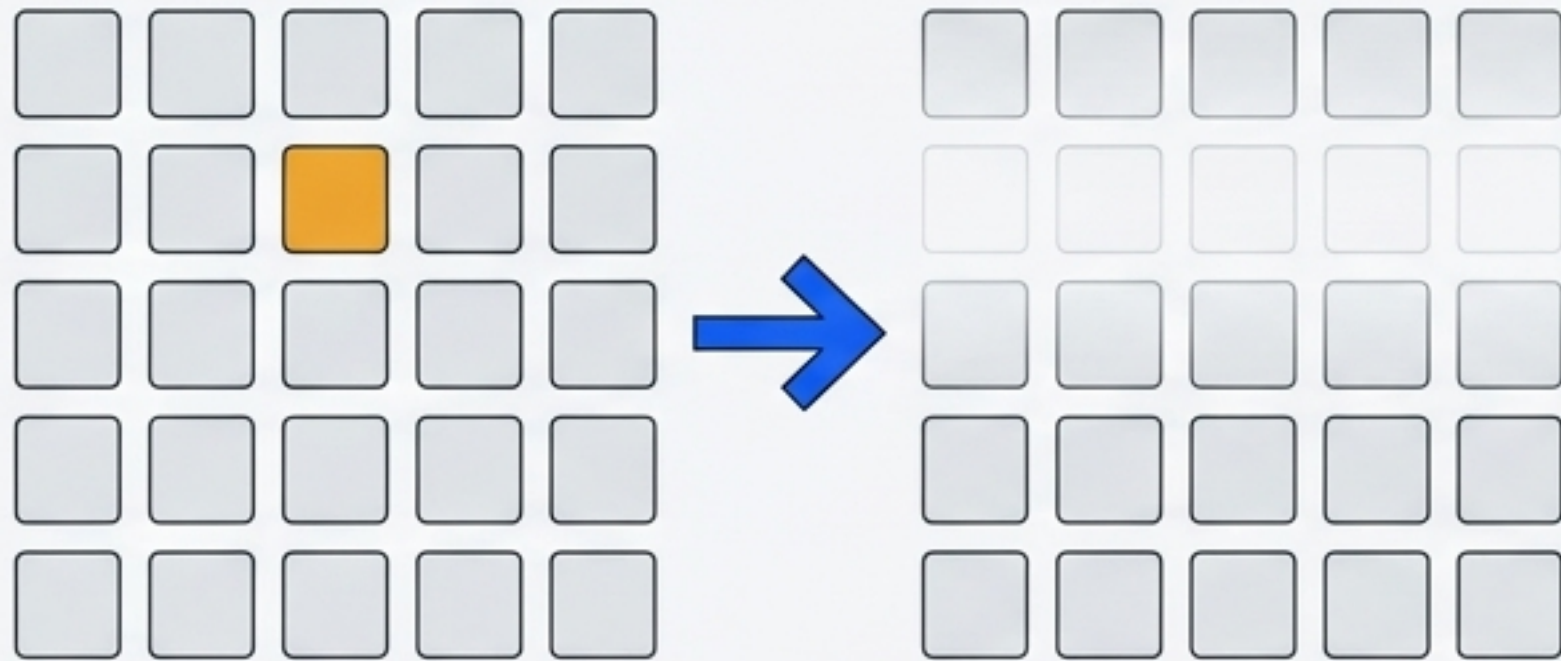


System Downtime: For online systems like chatbots, training data is volatile and full retraining causes service disruption.

Data Availability: Privacy regulations may prohibit keeping the original data indefinitely, making retraining legally impossible.

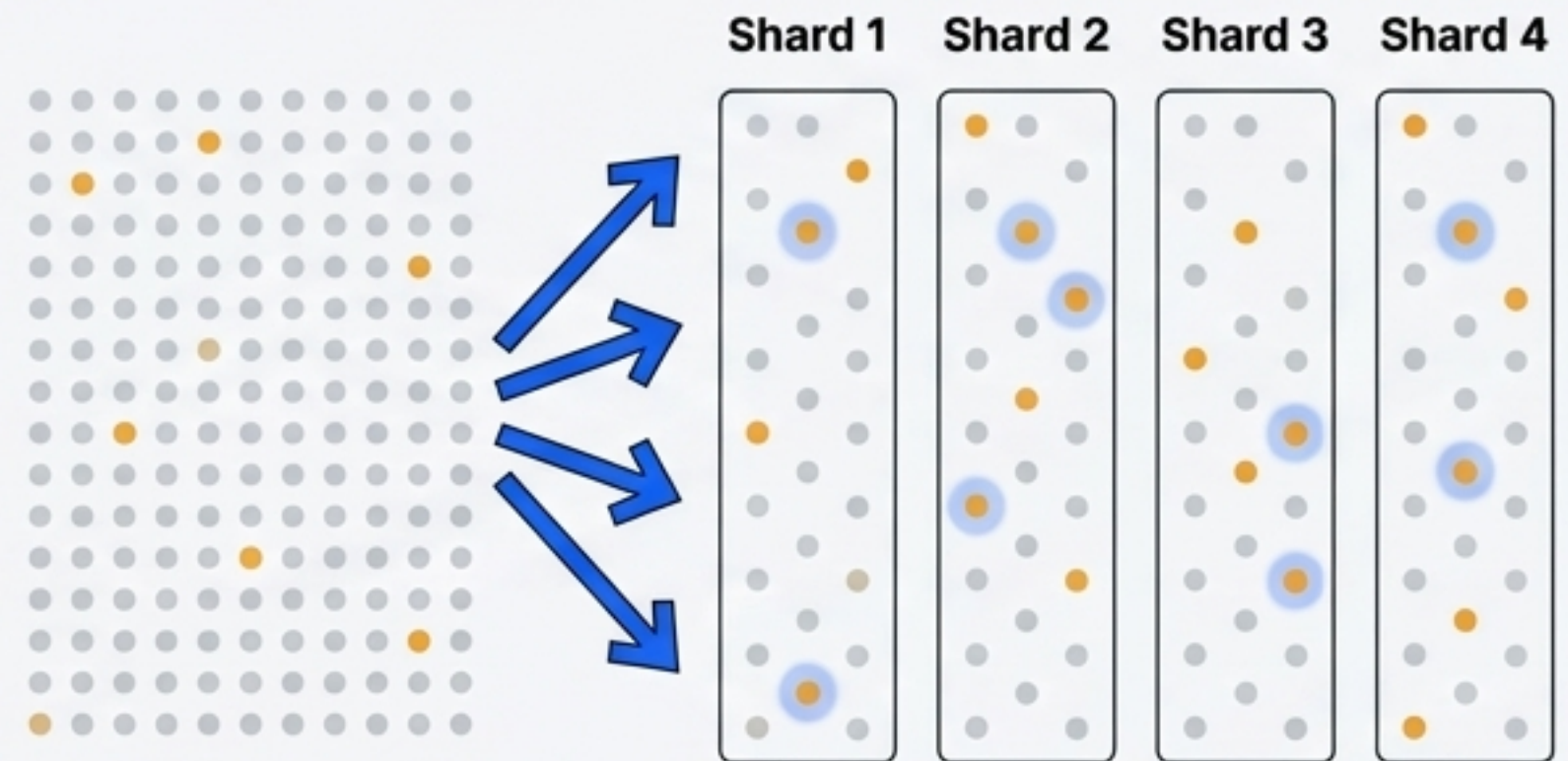
Existing 'shortcuts' are blunt instruments, not scalpels.

Instance-Based Unlearning



Too Blunt: To correct a single feature, an entire data point must be removed, sacrificing valuable information.

Sharding

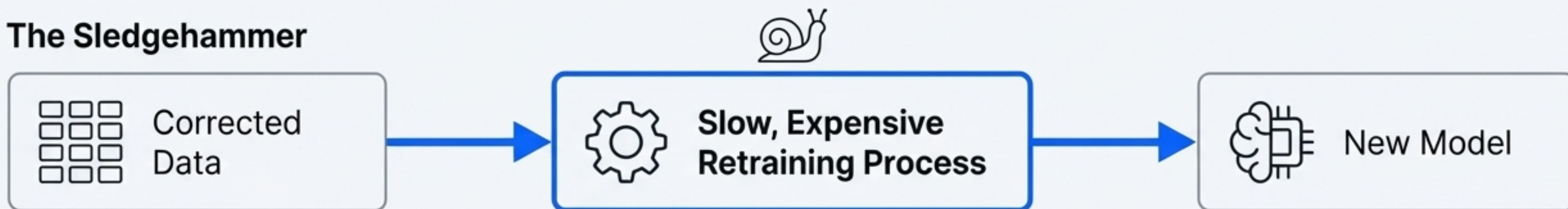


Breaks Down at Scale: Sensitive data can quickly spread across all partitions. With 20 shards, **just 150 data points needing** correction can force a 100% retrain of all shards. (Source: Figure 1).

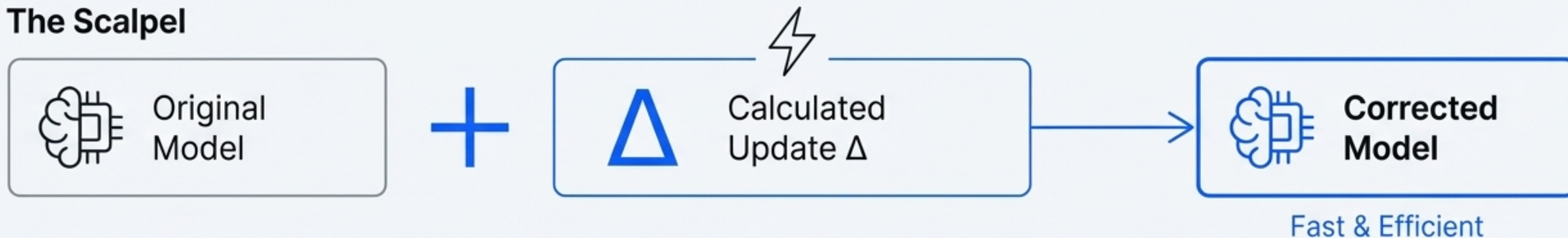
A new paradigm: Instead of retraining the data, we update the model.

Our approach avoids retraining by calculating a precise, closed-form update (Δ) that directly modifies the model's parameters. This is made possible by leveraging *influence functions*—a technique from robust statistics that estimates how much each piece of data influenced the final model parameters.

The Sledgehammer



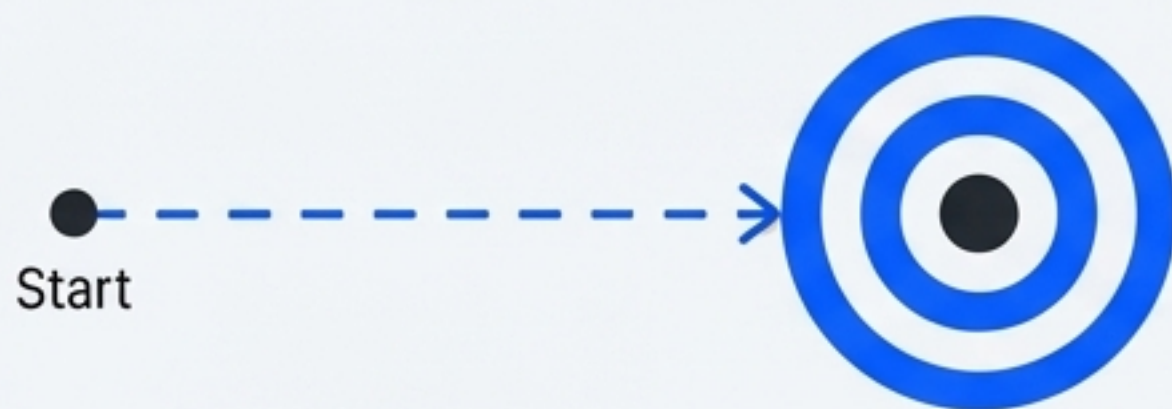
The Scalpel



Two paths to precise unlearning.

We derive two strategies for calculating the model update, offering a trade-off between speed and precision.

First-Order Update



A fast, direct update based on the data's gradient. It moves the model parameters away from the influence of the original data and towards the corrected version.



Analogy: "Like a single, smart step of gradient descent."



Pros: Extremely fast; can be orders of magnitude faster than retraining.



Cons: Requires calibration of an "unlearning rate" (τ).

Second-Order Update



A more precise update that also considers the loss "curvature" (the Hessian). This finds a more optimal path for the parameter update.



Analogy: "Finds the most direct path to the corrected state."



Pros: More accurate and parameter-free. Can provide theoretical guarantees.



Cons: More compute-intensive due to the inverse Hessian calculation.



Beyond 'Probably Gone': Mathematical Guarantees of Removal.

For models with strongly convex loss functions (e.g., Logistic Regression), our second-order update enables **certified unlearning**.

Definition

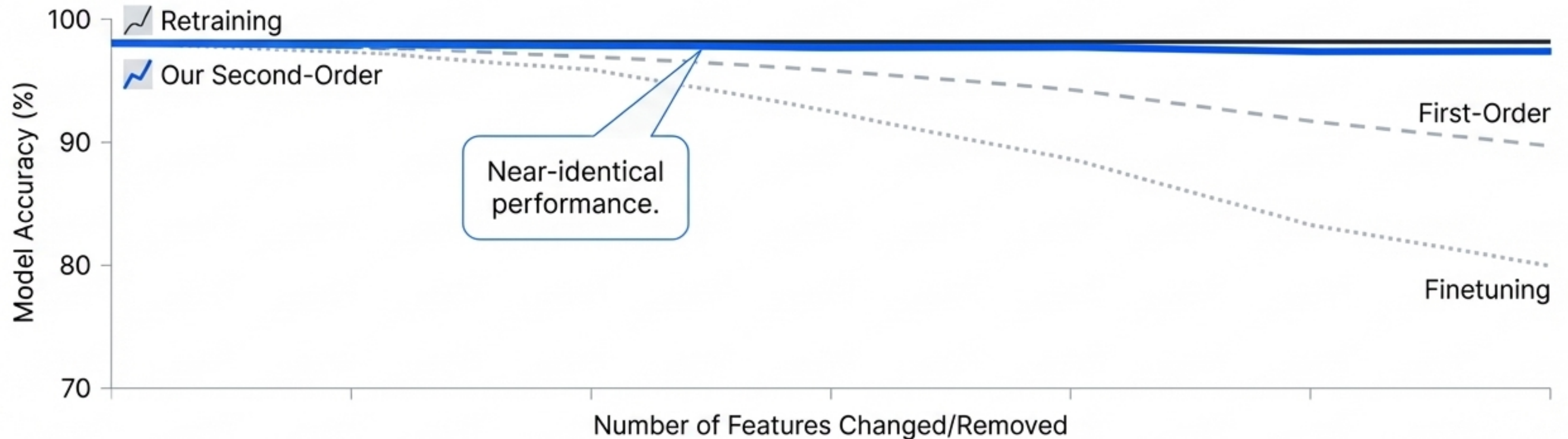
We introduce the concept of **ϵ -certified unlearning**. In simple terms, this means the unlearned model is provably indistinguishable from a model that was fully retrained from scratch on the corrected data. The probability of obtaining a given model⁷ our update method versus retraining deviates by at most a factor of **ϵ** .

Case Study 1: Excising Sensitive Features from Classifiers

Task: Removing potentially discriminatory features (e.g., race, sex) from linear models trained on datasets like Adult Census and Malware detection. Inter Regular

Key Metric: Fidelity (Test Accuracy). How well does the model perform after unlearning? Inter Regular

Unlearning maintains high accuracy, mimicking a full retrain.



Our second-order update provides **certified unlearning with **almost no loss in model performance**, closely matching the gold standard of retraining.**

Case Study 2: Forgetting a Secret in a Language Model

Task: A 'canary' (a fake phone number) was inserted into the training text of an LSTM. The task is to make the model 'forget' this number.

Key Metric: Efficacy (Exposure). How likely is the model to reveal the secret? Lower is better.

Our method eliminates the secret; fine-tuning does not.



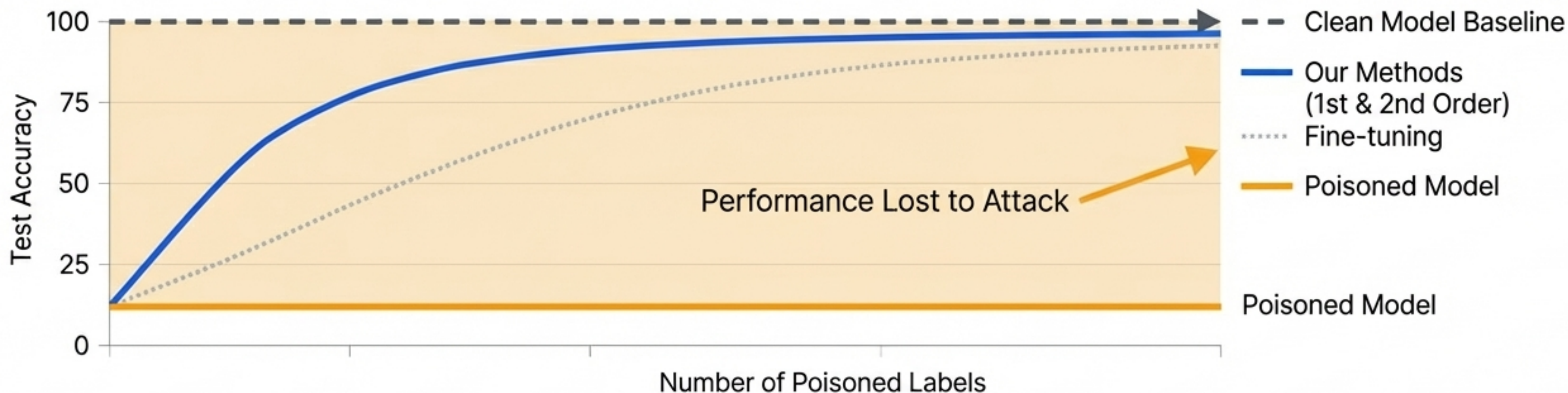
Our approach successfully makes the secret impossible to extract, where simpler methods like fine-tuning completely fail.

Case Study 3: Repairing a Poisoned Image Classifier

Task: An adversary flips labels in the CIFAR-10 training data (e.g., labeling 'cats' as 'trucks') to degrade model performance. The goal is to correct these labels and restore accuracy.

Key Metric: Accuracy Recovery. How close can we get back to the original, clean model's performance?

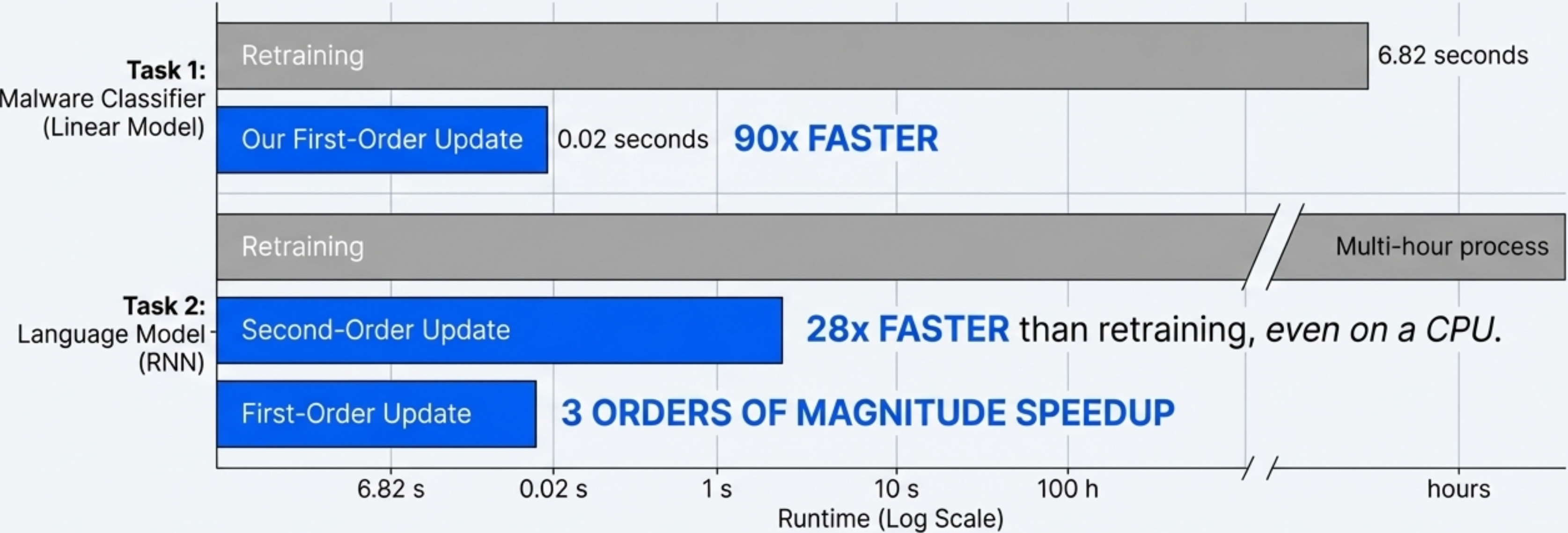
Unlearning quickly repairs damage from label poisoning attacks.



Unlearning is a powerful tool for **security**, enabling **rapid repair** of data poisoning attacks without the cost of a full retrain.

Precision with Speed: Unlearning in Seconds, Not Hours.

Across all scenarios, our influence-based updates provide massive speedups over retraining, turning a costly, multi-hour process into a near-instantaneous correction.



This level of **efficiency** makes **proactive and regular model maintenance feasible** for the first time.

A New Toolkit for Responsible and Maintainable AI.



EFFICIENT

Orders of magnitude faster than retraining, enabling rapid correction of data issues. Unlearning takes seconds, not hours.



FLEXIBLE

A single, unified framework to unlearn features, correct labels, revoke data, or remove entire instances as needed.



TRUSTWORTHY

The first method to provide `ε-certified` unlearning for features and labels, offering mathematical guarantees of removal for convex models.

Start Building More Responsible AI Today.

Dive deeper into the research and experiment with the unlearning framework yourself.
All implementations and datasets are publicly available.

Read the Full Paper



ndss-symposium.org/ndss2023/ndss-2023-paper-180/

Get the Code



github.com/alewarne/MachineUnlearning

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